

PROFESSIONAL SUMMARY

Researcher focused on the post-training and evaluation of frontier LLMs. Passionate about building tasks and environments that reveal the gaps, misalignment, and failure modes of LLMs, and building RL algorithms, feedback mechanisms, data flywheels, and human supervision pipelines to make models better for humanity.

EDUCATION

- **Carnegie Mellon University, School of Computer Science** Pittsburgh, PA
M.S. in Intelligent Information Systems (3.84/4.00) Aug 2022 - May 2024
- **Georgia Institute of Technology, Emory University, Dual Degree Program** Atlanta, GA
B.S in Computer Science, B.A. in Math and Economics (3.92/4.00) Aug 2017 - May 2022

RELEVANT PUBLICATIONS

* denotes equal contribution

- **RIFT: A Rubric Failure Mode Taxonomy and Automated Diagnostics:** (Zhengyang Qi, Charles Dickens, Derek Pham, Amanda Dsouza, Armin Parchami, Fred Sala, Paroma Varma) *ICLR 2026 Data Problems for Foundation Models Workshop*.
- **BeTaL: Adaptive Benchmarking with LLM-Guided Environment Design:** (Amanda Dsouza, Harit Vishwakarma, Zhengyang Qi, Justin Bauer, Derek Pham, Tom Walshe, Armin Parchami, Fred Sala, Paroma Varma) *ICML 2026. Under review*.
- **Sotopia-RL: Reward Design For Social Intelligence:** (Zhengyang Qi*, Haofei Yu*, Yining Zhao*, Keyang Xuan, Kolby Nottingham, Bodhisattwa Prasad Majumder, Hao Zhu, Paul Pu Liang, Jiaxuan You) *ICML 2026. Under review*.
- **MMoE: Enhancing Multimodal Models with Mixtures of Multimodal Interaction Experts:** (Haofei Yu*, Zhengyang Qi*, Lawrence Jang*, Ruslan Salakhutdinov, Louis-Philippe Morency, Paul Pu Liang) *The 2024 Conference of EMNLP. Main Conference*.
- **Sotopia- π : Interactive Learning of Socially Intelligent Language Agents:** (Ruiyi Wang*, Haofei Yu*, Wenxin Zhang*, Zhengyang Qi*, Maarten Sap, Graham Neubig, Yonatan Bisk, Hao Zhu) *The ACL 2024 Conference. Main Conference*.
- **Sotopia: Interactive Evaluation for Social Intelligence in Language Agents:** (Xuhui Zhu*, Hao Zhu*, Leena Marthur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, Maarten Sap) *The ICLR 2024 Conference. Spotlight*.
- **Long-Horizon Dialogue State Tracking for Role Identification in the Game of Avalon with Large Language Models:** Simon Stepputtis, Joseph Campbell, Yaqi Xie, Zhengyang Qi, Wenxin Zhang, Ruiyi Wang, Sanketh Rangreji, Charles Michael Lewis, Katia P. Sycara *Findings of the EMNLP 2023 Conference*.

WORK EXPERIENCES

- **Snorkel AI** San Francisco, CA
Research Scientist Sep 2025 – Present
 - **RL Post-Training and Environment Curation for Terminal Coding Agents:** Led post-training of LLM-based terminal agents on internal **Terminal Bench** tasks via **GRPO**. Curated terminal tasks by difficulty and scaled synthetic **mid-training** trajectory generation for diverse multi-step reasoning behaviors with **rejection sampling**. Built **multiturn RL** training pipelines using **Ray**-based distributed frameworks with **async rollouts** and **LLM-as-Judge** rewards. Trained **Qwen-3-8B** and **Qwen-3-32B**, improving task success **2.2x** (5.6% to 13.4%).
 - **Research Agentic RL Systems:** Built RL systems from scratch from verl. This includes cluster management, adaptation to Ray based training frameworks (Verl, SkyRL, etc) ray and kubernetes cluster management on aws, file system and storage system agentic rollout management for docker images, resulting in robust distributed training pipelines spanning 64 H200 GPUs.
 - **Professional Benchmark and Environment for Computer-Use Agents:** Designed and built **50+** interactive tasks and environments for visual computer-use agents, creating a robust evaluation suite for complex interface grounding, diverse action tool use, and long-context and test-time compute behaviors in professional circuit design tasks.
 - **Rubrics and Synthetic Data for Medical Agents:** Designed and supervised rubric-based and preference-based human annotation pipelines for medical conversational agents. Prototyped **long-context (40k+)** synthetic data generation via EHR simulation to support training and evaluation of clinical reasoning models.
- **Scale AI** San Francisco, CA
Machine Learning Research Engineer Aug 2024 - Aug 2025
 - **Post-Training of LLM Judge Models for Reward & Preference:** Led post-training for LLM judge models using **SFT** and **GRPO** to align human reward and preference. Built scalable production evaluation platform for reward models and judge agents, and curated **60k+** comprehensive datasets for frontier capabilities (multilingual instruction following, reasoning, STEM, and coding). Experimented with open-weights models including **Llama 3.3 70B**, **Qwen2.5**, **QwQ 32B**, optimizing training recipes via data selection, augmentation, and RL reasoning to overcome reward hacking and spurious alignment. Delivered a SOTA production reward judge model **outperforming GPT-4o** in reward and preference alignment.
 - **Scalable Oversight & Quality Control Agentic Systems:** Led **scalable oversight** initiatives for human-AI data generation pipelines. Built a production agentic auto-evaluation system via multi-agent debate and context management and developed reliable metrics for system effectiveness through **A/B tests**. Deployed top **10+** evaluation pipelines, dramatically improving data quality and achieving up to **60%** reduction in human labeling reliance.

- **Rubrics RLHF on Multimodal Expert Tasks:** Developed automated quality control agents for RLHF Rubrics pipelines via RAG, tool use, and multi-agent debate. Demonstrated consistent evaluation improvements across deployments over baseline heuristics.

• Intuit

San Diego, CA

May - Aug 2023

Software Engineer (Machine Learning) Intern

- **Tax Form De-Duplication with LLMs:** Designed and implemented **scalable** machine learning microservices to identify redundant data across **1k+** variations of tax forms for **42M users**, integrating **GPT-3.5**, **XGBoost**, and **Spring Boot** to automate data curation and feature extraction.

• Intuit

Remote

Jun - Aug 2022

Software Engineer (Machine Learning) Intern

- **Multimodal Language Model & Inference Optimization:** Constructed image captioning model utilizing SOTA representations (**GPT2/3**, **CLIP**) performing **43%** over baseline. Created a **Spark** pipeline for **70k+** images; pruned model size by **68%**; deployed batch inference via **CUDA** and **PyTorch FSDP**, reducing inference latency by **82%**.

• Kiwi

Atlanta, GA

Dec 2020 - Dec 2021

Co-Founder & Machine Learning Engineer

- **ML & Data Pipeline for Recommender System:** Led a 3-engineer team building a multi-modal recommender system across 18 retailers. Automated acquisition of **100k+ items**; experimented with **XGBoost/LGBM** for entity matching (**F1=.89**), achieving a **64%** improvement over baseline.

PROJECTS

- **Chinese FLAN-T5 (Open Source):** Contributed **20+** pipelines for instruction following LLMs in Chinese and 8 languages.
- **Amex Default Prediction (Kaggle Top 1.6%):** Experimented with ensemble models and advanced feature engineering.
- **Distributed Systems (Go):** Implemented fundamental distributed paradigms including **UDP-based** reliable network protocols, **RAFT consensus** algorithm, and the **Actor model** for multi-user state management.
- **Search Engine Systems (Java, Lucene):** Developed an OOP-centric search-engine library implementing comprehensive **IR** algorithms including **BM25**, **Indri**, Learning-to-Rank, and diversification.

TECHNICAL SKILLS

- **Languages:** Python, C++, Go, Java, TypeScript, SQL, Unix shell
- **ML & Systems:** PyTorch, Ray, vLLM, Triton, CUDA, Distributed Training (FSDP, DeepSpeed), Spark
- **Frontier AI:** RLHF/RLAIF (PPO, GRPO, DPO, ReST), Rubrics, Reward Modeling, Scalable Oversight, Synthetic Data Generation, Test-Time Compute, Chain-of-Thought (CoT), Multimodal Models